

Hind Zabrati

Engineering Student in IT Security

Seeking a 6-month end-of-studies internship (PFE) – February 2027

Offensive & Defensive Security | Automotive Security | LLM

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PROFILE

Final-year engineering student in IT Security, trained in offensive and defensive security (Active Directory and network pentesting, intrusion detection), with a strong interest in automotive security and LLM applications. I am looking for a 6-month end-of-studies internship (PFE) starting February 2027.

EXPERIENCE

Cybersecurity Intern – Automotive Security — **July – August 2026**

Ovivia AI Solution — *Doha, Qatar*

- Designed automotive risk scenarios (ISO/SAE 21434, UNECE R155/R156) using a standardized methodology
- Deployed a multi-VM environment (Proxmox VE) and built a 9-task CAN Bus Speed Spoofing challenge (MITRE ATT&CK for ICS)
- Proposed the platform's content architecture, now used to structure all scenarios

Cybersecurity Intern – Attack Simulation and Detection — **July 2025**

Ménara Holding — *Marrakech*

- Simulated an attack chain (Tomcat → reverse shell → privilege escalation) on a Kali/Metasploitable 2 lab
- Deployed Suricata and Wazuh with custom IDS rules, analyzed alerts and identified IoCs

PROJECTS

CabinetShield – Open-Source IDS Appliance for Medical Practices — **Ongoing**

Suricata, Python, Flask, SQLite, REST API, Debian

- Local appliance built in a pair: Suricata as a passive IDS (mirror port) and a Python collector (normalization, EVE JSON deduplication)
- Authenticated REST API (Flask, SQLite) and web dashboard explaining alerts in plain French

Audit and Hardening of an LLM Application

OWASP Top 10 for LLM, Ollama, ChromaDB, Gradio

- Gray-box audit of an internal assistant (RAG, tool calls): prompt injection, data leakage, excessive agency, with success rates measured over multiple runs
- Remediation (filters, RAG access control, tool validation) and before/after re-testing, with monitoring recommendations

Full Active Directory Domain Compromise

BloodHound, Kerberos, Mimikatz, Impacket

- Reconnaissance (BloodHound), password spraying and Kerberos escalation (Kerberoasting, Golden Ticket via DCSync)
- Post-exploitation: NTDS.dit dump, lateral movement (Pass-the-Hash) and remediation recommendations

Design and Hardening of an Enterprise Architecture

OpenVPN, pfSense, Suricata

- Segmented architecture (pfSense, Ubuntu, Kali) with Client-to-Site VPN (OpenVPN, internal PKI) and Ubuntu hardening
- Deployed Suricata as an inline IPS, validated through simulated network attacks (scans, brute force, flood)

EDUCATION

Engineering Degree in IT Security — **2024 – 2027**

Faculty of Sciences and Techniques of Marrakech – Cadi Ayyad University

DEUST in MIPC (Mathematics, Computer Science, Physics, Chemistry) — **2021 – 2024**

Faculty of Sciences and Techniques of Marrakech – Cadi Ayyad University

TECHNICAL SKILLS

Offensive Security	Nmap, Metasploit, BloodHound, Mimikatz, NetExec, Impacket, Hydra, Hashcat	**Defensive Security**	Wazuh (SIEM), Security Onion, Suricata (IDS/IPS), Wireshark
LLM Security	OWASP Top 10 for LLM, RAG, Ollama, ChromaDB, garak, LLM Guard	**Development & Data**	Flask, REST API, SQLite, Docker Compose, Debian
OT/ICS & Automotive	CAN Bus, ICSim, Proxmox VE, TARA, automotive security standards	**Networking**	pfSense, OpenVPN, GNS3
Cryptography & Governance	AES, RSA, Diffie-Hellman, SHA-256, ISO 27001, risk analysis	**Systems & Programming**	Linux, Windows, Active Directory, Python, C, Java, JavaScript, PHP, SQL

CTF COMPETITIONS

MACC 2026 (DGSSI & SecDojo) — **1st place** at FSTG and **62nd** out of 1068 teams worldwide

ST4F1T CTF (ENSA Marrakech) — **8th place** – Forensics, OSINT, Reverse Engineering, Web, Cryptography

CERTIFICATIONS

AI Security Path – TryHackMe

DevSecOps Path – TryHackMe *(ongoing)*

Cybersecurity 101 Path – TryHackMe

Ethical Hacker – Cisco

Introduction to Cybersecurity – Cisco

LANGUAGES

French : fluent

English : fluent